
Exercise 1: PDF report and documentation

Abstract

This report presents a statistical comparison of three datasets provided by the professor. The objective of the analysis is to determine if the datasets originate from the same statistical distributions or from different ones by applying the hypothesis tests studied in the theory classes with Francisco Carrera. Comparisons of the datasets are performed in pairs at a 95% confidence level using the F-test, F-test with ranks, Student's t -test, Wilcoxon rank-sum test (Student's t -test with ranks), and the two-sample Kolmogorov-Smirnov (KS) test.

The main findings are that the F-test and Student's t -test do not detect differences in variances or means (p -values ≈ 1), while rank-based tests identify significant differences for several pairs. The two-sample KS test strongly rejects the null hypothesis of identical distributions for all three pairs (p -values ≈ 0). Then the overall conclusion is that despite having similar means, the three samples come from different distributions as the KS test suggests and as histograms and boxplots confirm.

1 Introduction

The three datasets analysed in this report were provided by the professor Maite Ceballos the class of the 28th of October of 2025 as part of the course assignment. According to their numerical properties, they appear to have been selected in such a way that some of their basic descriptive statistics, such as the mean or the median, are similar, while other aspects of their distributions differ. The data has just been readen from the provided `.dat` files without any further transformations, and all hypothesis tests are carried out directly on the original values.

The objective of this study is to determine whether the datasets come from similar or different (statistically) populations. Several hypothesis tests are performed in pairs: the F-test for the equality of variances [2], the F-test with ranks [3], the Student's t -test [2], the Student's t -test with ranks (Wilcoxon) [3], and the two-sample Kolmogorov-Smirnov (KS) test [2] using the R [1] language. These tests compare variances, means, medians, and full distributional shapes, allowing the comparison of the datasets from multiple statistical perspectives.

All tests are performed using a significance level of $\alpha = 0.05$. For each comparison, the null hypothesis corresponds to the assumption that both samples come from populations with the

same corresponding statistical property (equal variances, equal means, equal median/rank distributions, or identical distribution functions). When the p -value is below the significance level, which is of 95% confidence ($\alpha = 0.05$), the null hypothesis is rejected.

2 Results

2.1 Preliminary summary statistics

Before performing the hypothesis tests, compute some preliminary summary statistics are computed for each dataset with the function `summary()` of R [1]. The results are summarised in the following table:

Statistic	<code>dat1</code>	<code>dat2</code>	<code>dat3</code>
Min.	-0.463	0.000	-3.990
1st Qu.	0.747	1.000	1.023
Median	1.976	2.000	2.002
Mean	1.986	1.986	1.986
3rd Qu.	3.226	3.000	2.936
Max.	4.443	9.000	8.239

Table 1: Summary statistics for the three datasets (`dat1`, `dat2`, `dat3`). The table includes the minimum, first quartile, median, mean, third quartile, and maximum values for each dataset.

From this first overview, and as previously mentioned in the introduction, all three datasets have similar means (approximately 1.986) and close medians (from 1.976 to 2.002).

2.2 Visual inspection of the datasets

Before performing the statistical tests, two graphical summaries can be produced to visualise the datasets in order to spot potential differences in distributional shape, central tendency, and dispersion. These figures also provide preliminary intuition and guide the interpretation of the formal hypothesis tests.

The first figure presented is a set of histograms, one for each dataset:

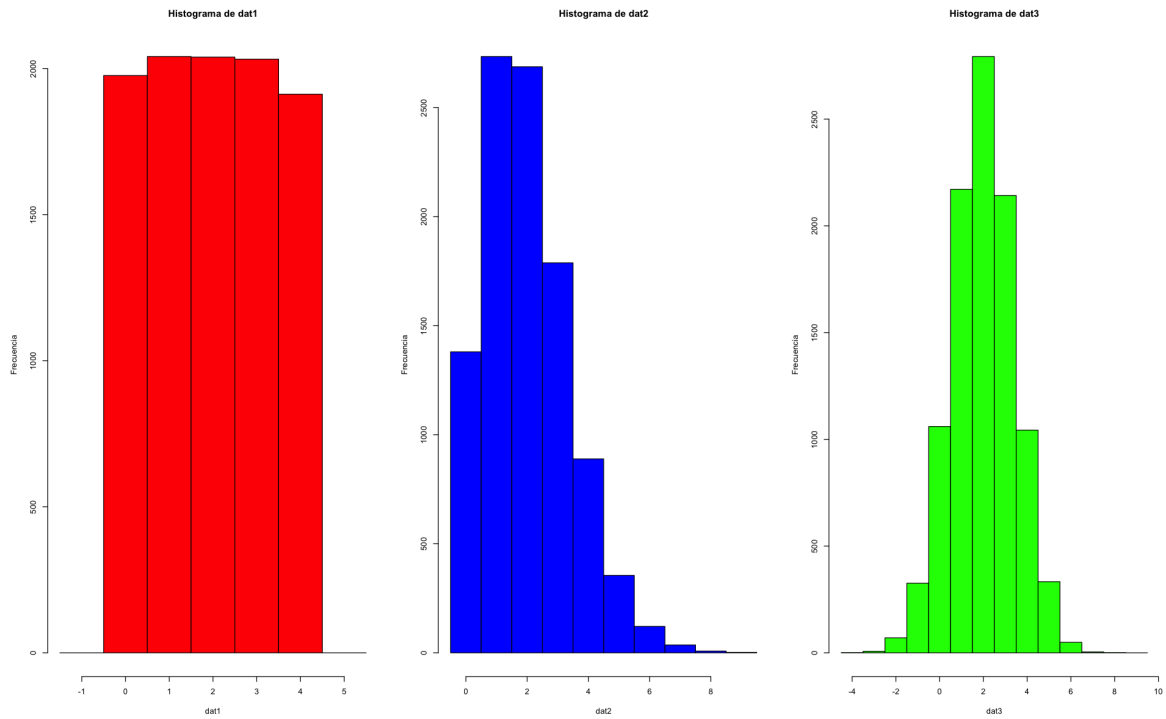


Figure 1: Comparison of the histograms of the three datasets. Each colour corresponds to one dataset labeled accordingly. The size of the bins is chosen to be unitary for clarity.

The second figure is a set of boxplots, again one for each dataset:

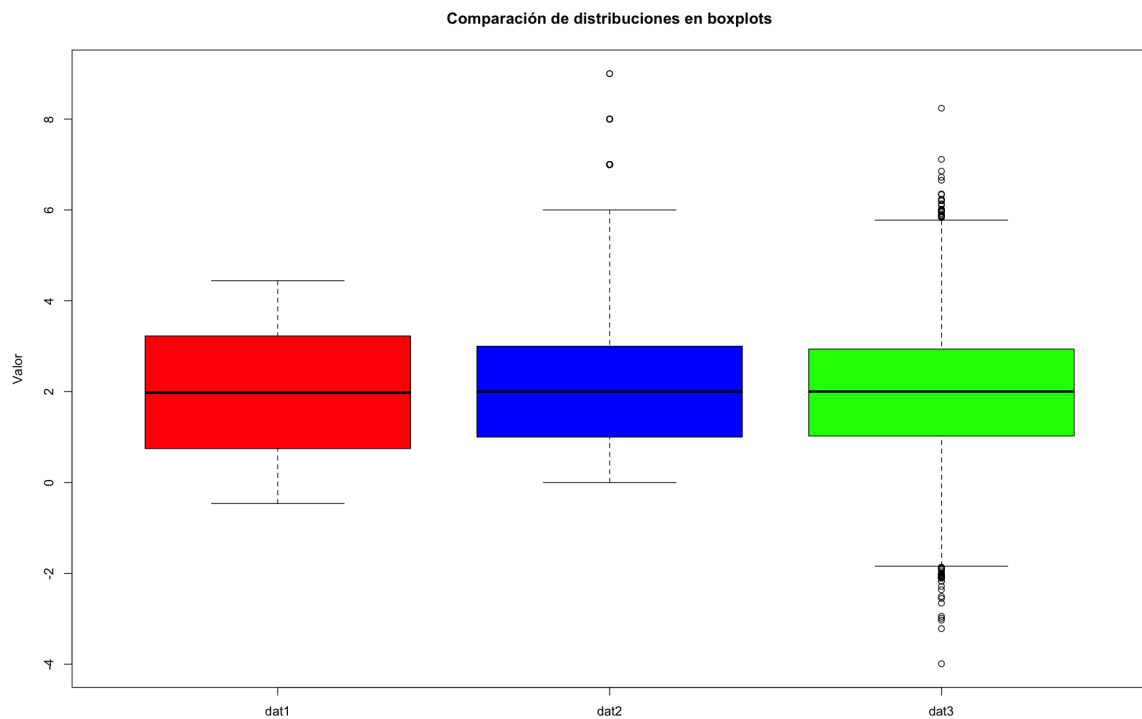


Figure 2: Comparison of the boxplots of the three datasets. Each colour corresponds to one dataset, in agreement with the histograms of Figure 1. The boxplots display the median, interquartile range, and potential outliers for each dataset.

The histogram comparison in Figure 1 suggests that although the datasets may share similar means, their overall shapes differ substantially, particularly in dispersion and tail behaviour. The boxplots in Figure 2 confirm that the datasets exhibit different interquartile ranges and distributional asymmetries, which motivates the use of both parametric and non-parametric (with ranks) tests.

2.3 Summary of p -values from hypothesis tests

The analysis consists of applying five different hypothesis tests to the three datasets provided: the F-test, the F-test with ranks, the Student's t -test, the Student's t -test with ranks (Wilcoxon), and the two-sample Kolmogorov-Smirnov (KS) test. All of the mentioned tests are performed pairwise among the datasets (`dat1`, `dat2` and `dat3`) and evaluated at the 95% significance level.

All the p -values obtained from each test to each pair of data is summarised in the next table:

Test	dat1 - dat2	dat1 - dat3	dat2 - dat3
F-test	1.000	1.000	1.000
F-test with ranks	$< 2.2 \times 10^{-16}$	1.821×10^{-14}	0.017
Student's t -test	1.000	1.000	1.000
t -test with ranks (Wilcoxon)	0.010	0.851	2.959×10^{-5}
Kolmogorov-Smirnov (KS)	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$

Table 2: Summary of the p -values obtained from all hypothesis tests applied to the three datasets (`dat1`, `dat2`, `dat3`). Each column corresponds to a dataset comparison, and each row to a specific statistical test. All tests were performed two-sided at a significance level of $\alpha = 0.05$. A p -value below this threshold indicates rejection of the corresponding null hypothesis.

The discussion of the results for each test follows in the subsequent subsections.

2.3.1 Comparison of variances: F-test

The p -values of the F-test for the equality of variances are located in the first row of Table 2 and are extremely close to 1, indicating no evidence of different variances between any pair of datasets. Therefore, at the 95% confidence level, the null hypothesis of equal variances cannot be rejected for any pair.

2.3.2 Comparison of rank distributions of variances: F-test with ranks

The p -values of the rank-based F-test, shown in the second row of Table 2, reveal a different picture. Two of the three comparisons yield p -values equal to zero (within numerical precision), indicating a very strong rejection of the null hypothesis of equal rank distributions. The third

comparison (`dat2-dat3`) yields a p -value still below the threshold ($p = 0.017$), so the null hypothesis can still be rejected at $\alpha = 0.05$ but with less confidence.

2.3.3 Comparison of means: Student's t -test

The results of the Student's t -test on p -values, given in the third row of Table 2, which show no statistical evidence of differences in the means of the datasets. All of them are virtually 1, which agrees with the previous observation that the datasets were selected so that their means are very similar.

2.3.4 Comparison of medians and rank distributions: Student's t -test with ranks (Wilcoxon)

The fourth row of Table 2 reports the p -values of the Wilcoxon rank-sum test. Two of the three comparisons yield extremely small values (`dat1-dat2` and `dat2-dat3`), rejecting the null hypothesis of equal median distributions. However, the comparison between `dat1` and `dat3` produces a large value ($p = 0.851$) close to 1, so that the null hypothesis of equal median distributions cannot be rejected for this pair at the 95% confidence level. Then, the pair `dat1-dat3` appears to share similar median properties, while the other two pairs differ significantly.

2.3.5 Comparison of full distributions: Kolmogorov–Smirnov test

Finally, the last (fifth) row of Table 2 summarises the p -values of the two-sample Kolmogorov–Smirnov test. All of them are virtually zero (below computational limits), providing evidence against the null hypothesis that the pairs of datasets originate from the same underlying distribution. This test, being sensitive to the entire cumulative distribution function, confirms that the datasets are fundamentally different in shape, regardless of similarities in mean or variance. This is in strong agreement with the visual inspection of histograms and boxplots in Figures 1 and 2.

3 Conclusions

This study compared three provided samples (`dat1`, `dat2`, `dat3`) taking advantage of the R [1] language by applying several hypothesis tests to determine whether they originate from the same statistical distribution.

The data were analysed without any transformation using five pairwise hypothesis tests: the F-test (variance), F-test on ranks, Student's t -test (mean), Wilcoxon rank-sum test (ranks/medians), and the two-sample Kolmogorov–Smirnov (KS) test. Preliminary results are summarised in Table 1, followed by the visualization of the data via histograms and boxplots in Figures 1 and 2. Finally, the p -values from all tests are compiled in Table 2.

The parametric tests for variance and mean (F-test and Student's t -test) produce very large p -values and do not reject equality of variances or means for any pair. In contrast, rank-based tests (F-test with ranks and Wilcoxon) identify significant differences for several pairings. The KS test rejects equality of distributions for all three pairs with extremely small p -values, as expected from the visual inspection of the data previously done with histograms and boxplots of Figures 1 and 2. The combination of all these results justifies treating the samples as coming from different populations.

References

- [1] R Core Team, *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, 2025. <https://www.R-project.org/>
- [2] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, Cambridge, 2007. ISBN 978-0-521-88068-8.
- [3] Sheskin, D. J., *Handbook of Parametric and Nonparametric Statistical Procedures*, 5th edition, Chapman and Hall/CRC, Boca Raton, 2011. ISBN 978-1-4398-4093-2.